

A Few Options For Sampling Window Plots

Eleanor (Ella) Crotty

Setup

```
# Packages
library(tidyverse)

## Warning: package 'ggplot2' was built under R version 4.3.3
## Warning: package 'tibble' was built under R version 4.3.3
## Warning: package 'tidyr' was built under R version 4.3.1
## Warning: package 'purrr' was built under R version 4.3.3
## Warning: package 'dplyr' was built under R version 4.3.1
## Warning: package 'lubridate' was built under R version 4.3.3
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2    3.5.2      v tibble     3.3.0
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
library(patchwork)
library(scales)

## Warning: package 'scales' was built under R version 4.3.3
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##   discard
##
## The following object is masked from 'package:readr':
##
##   col_factor
library(ggrepel)

## Warning: package 'ggrepel' was built under R version 4.3.3
```

```
library(ggbreak)
```

```
## ggbreak v0.1.6 Learn more at https://yulab-smu.top/
##
##
## If you use ggbreak in published research, please cite the following
## paper:
##
## S Xu, M Chen, T Feng, L Zhan, L Zhou, G Yu. Use ggbreak to effectively
## utilize plotting space to deal with large datasets and outliers.
## Frontiers in Genetics. 2021, 12:774846. doi: 10.3389/fgene.2021.774846
```

```
library(here)
```

```
## here() starts at /Users/ellac/Dropbox/Coding/Summer2024/PMEL-Hollings-eDNA-Hypoxia-2024
```

```
# Import eDNA metadata
```

```
md <- read_csv(here("Manuscript_Plots/eDNA_Metadata.csv"))
```

```
## New names:
```

```
## * `Cruise/ Expedition ID` -> `Cruise/ Expedition ID...6`
```

```
## * `Cruise/ Expedition ID` -> `Cruise/ Expedition ID...7`
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
```

```
##   dat <- vroom(...)
```

```
##   problems(dat)
```

```
## Rows: 3407 Columns: 44
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (35): FINAL Sample NAME, Cruise/ Expedition ID...6, Cruise/ Expedition I...
```

```
## dbl (5): Sequential G3 Sample No., Biological Replicate No. (if applicable)...
```

```
## lgl (4): Negative control, Extracted, Sent for Sequencing, Sequencing compl...
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Filter for OCNMS samples
```

```
unique(md$`Cruise/ Expedition ID...7`)
```

```
## [1] "RR1712 Axial 2017"      "NA100 Loihi 2018"
## [3] "NA112 Vailulu'u 2019"  "NA117 MBNMS 2019"
## [5] "DY20-12 Dyson 2020"    "WOAC Sept 2020 (Micro)"
## [7] "TN383 Axial 2020"      "Norseman 2020"
## [9] "Carson Alki 2021"      NA
## [11] "WCOA 2021"             "WOAC Spring 2022"
## [13] "Cascadia 2021"         "OCNMS Aug 2021"
## [15] "SKQ21 EcoFOCI DB0"     "EcoFoci Spring Mooring 2022"
## [17] "OCNMS June 2022"       "WOAC Summer 2022"
## [19] "OCNMS August 2022"     "OCNMS Sept 2022"
## [21] "OCNMS Oct 2021"        "Axial 2022"
## [23] "OCNMS July 2022"       "RC Alaska OA Cruise 2022"
## [25] "WOAC Fall 2022"        "EcoFOCI Fall Mooring 2022"
## [27] "WOAC Cha'ba Cruise"    "RV Falkor"
## [29] "EcoFOCI Spring Mooring" "M2 Automated Sampler"
```

```
## [31] "Kotzebue Demo"          "OCNMS June 2023"
## [33] "OCNMS July 2023"        "SKQ23 EcoFOCI DB0"
## [35] "May 2024 PPS Deployment" "June 2024 PPS Recovery"
## [37] "June 2024 PPS Deployment"
```

```
cruise_na <- md %>% filter(is.na(`Cruise/ Expedition ID...7`)) # Some are basically blank
↳ rows, but some have the other ID
unique(md$`Cruise/ Expedition ID...6`)
```

```
## [1] "AX17"      "L018"      "VA19"      "DA19"      "DY2012"
## [6] "W0920"     "AX20"      "N020"      "RC56"      "RC50"
## [11] NA          "WCOA21"    "NA128"     "OC0821"    "SKQ2021"
## [16] "DY2206"    "OC0622"    "RC0080"    "OC0822"    "OC0922"
## [21] "OC1021"    "AX22"      "OC0722"    "RC0083"    "DY2209"
## [26] "DY2306"    "OC0623"    "OC0723"    "SKQ23-12S" "OC0919"
## [31] "QIAvac Test" "Kotzebue23" "Tonga2024" "DY2406"    "QiavacQuant"
## [36] "OC0524"    "OC0624"
```

```
md_oc <- md %>%
  filter(`Cruise/ Expedition ID...7` == "OCNMS August 2022" |
    `Cruise/ Expedition ID...7` == "June 2024 PPS Deployment" |
    `Cruise/ Expedition ID...7` == "OCNMS Aug 2021" |
    `Cruise/ Expedition ID...7` == "OCNMS June 2022" |
    `Cruise/ Expedition ID...7` == "OCNMS Sept 2022" |
    `Cruise/ Expedition ID...7` == "OCNMS July 2022" |
    `Cruise/ Expedition ID...7` == "OCNMS June 2023" |
    `Cruise/ Expedition ID...7` == "May 2024 PPS Deployment" |
    `Cruise/ Expedition ID...7` == "OCNMS Oct 2021" |
    `Cruise/ Expedition ID...7` == "OCNMS July 2023" |
    `Cruise/ Expedition ID...7` == "June 2024 PPS Recovery" |
    # Abbreviations
    `Cruise/ Expedition ID...6` == "OC0622" |
    `Cruise/ Expedition ID...6` == "OC0822" |
    `Cruise/ Expedition ID...6` == "OC0623" |
    `Cruise/ Expedition ID...6` == "OC0922" |
    `Cruise/ Expedition ID...6` == "OC0723" |
    `Cruise/ Expedition ID...6` == "OC0524" |
    `Cruise/ Expedition ID...6` == "OC1021" |
    `Cruise/ Expedition ID...6` == "OC0624" |
    `Cruise/ Expedition ID...6` == "OC0821" |
    # `Cruise/ Expedition ID...6` == "OC0919" | # Delete Chris Paight Samples
    `Cruise/ Expedition ID...6` == "OC0722"
  )
```

```
md_oc_pps <- md_oc %>%
  filter(`Sampling method` == "PPS Unit; Programmed Sampling")
unique(md_oc_pps$`Sampling method`)
```

```
## [1] "PPS Unit; Programmed Sampling"
```

```
unique(md_oc$`Sampling method`)
```

```
## [1] "Sterivex 0.22µm filter"
## [2] "Niskin; Sterivex 0.22µm filter"
## [3] "PPS Unit; Programmed Sampling"
```

```
## [4] "PPS Unit; Sterlitech 0.4 µm filter, 47 mm"
## [5] "Van Dorn Bottle; Sterivex 0.22µm filter"

sampledatelist <- as.POSIXct(md_oc_pps$`Collection Date (local)`,
                             tz = "America/Los_Angeles",
                             tryFormats = c("%m/%d/%Y"))

md_oc_ctd <- md_oc %>%
  filter(`Sampling method` != "PPS Unit; Programmed Sampling" &
         `Sampling method` != "PPS Unit; Sterlitech 0.4 µm filter, 47 mm")

ctddatelist <- as.POSIXct(md_oc_ctd$`Collection Date (local)`,
                          tz = "America/Los_Angeles",
                          tryFormats = c("%m/%d/%Y"))
```

Variables to change aesthetics

```
widthpx <- 4000 # Width for export (pixels)
heightpx <- 2000 # Height for export (pixels)

widthpx_stacked <- 2000 # Width for export (pixels) for the stacked plots
heightpx_stacked <- 2000 # Height for export (pixels) for the stacked plots

# Rectangle windows aesthetics
width = 1
linecolor = "black"
fillcolor = "navy"

# CTD line or dot aesthetics
ctdcolor = "firebrick4"
# Add CTD as either point or line - uncomment whatever one you want
CTD_add <- geom_point(aes(x = ctddatelist, y = 0), color = ctdcolor)
# CTD_add <- geom_vline(aes(xintercept = ctddatelist), color = ctdcolor) # add CTD

# Extract sampling windows manually - sort by collection date and take earliest & latest
↳ from each expedition ID
# Define highlight rectangles
# Will do more after getting updated metadata
sampleHighlight <- tibble(x1b = as.POSIXct("2022-06-23"), x1e = as.POSIXct("2022-07-19"),
  ↳ # OCNMS July 2022 - TH & CE
                        x2b = as.POSIXct("2021-08-26"), x2e = as.POSIXct("2021-09-29"),
  ↳ # OCNMS Oct 2021 - TH & CE
                        x3b = as.POSIXct("2023-06-16"), x3e = as.POSIXct("2023-07-20"),
  ↳ # OCNMS July 2023 - TH
                        x4b = as.POSIXct("2022-08-23"), x4e = as.POSIXct("2022-09-18"),
  ↳ # OCNMS Sept 2022 - TH
                        y1 = -Inf, y2 = +Inf) # INF IS NECESSARY TO KEEP THE RECTANGLES
  ↳ RUNNING OFF THE PAGE

SampHighlight1 <- geom_rect(data = sampleHighlight,
                           inherit.aes = FALSE,
                           mapping = aes(xmin = x1b, xmax = x1e,
                                          ymin = y1, ymax = y2),
```

```

        color = linecolor,
        fill = fillcolor,
        linewidth = width,
        alpha = 0.2)
SampHighlight2 <- geom_rect(data = sampleHighlight,
        inherit.aes = FALSE,
        mapping = aes(xmin = x2b, xmax = x2e,
                      ymin = y1, ymax = y2),
        color = linecolor,
        fill = fillcolor,
        linewidth = width,
        alpha = 0.2)
SampHighlight3 <- geom_rect(data = sampleHighlight,
        inherit.aes = FALSE,
        mapping = aes(xmin = x3b, xmax = x3e,
                      ymin = y1, ymax = y2),
        color = linecolor,
        fill = fillcolor,
        linewidth = width,
        alpha = 0.2)
SampHighlight4 <- geom_rect(data = sampleHighlight,
        inherit.aes = FALSE,
        mapping = aes(xmin = x4b, xmax = x4e,
                      ymin = y1, ymax = y2),
        color = linecolor,
        fill = fillcolor,
        linewidth = width,
        alpha = 0.2)

```

geom_rect

```

# Graph 1: Shaded gray sampling windows

ggplot() +
  SampHighlight1 +
  SampHighlight2 +
  SampHighlight3 +
  SampHighlight4 +
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 12),
        strip.background = element_rect(fill = "gray95"), # facet label background
        axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
        ↪ ggbreak
        axis.ticks.x.top = element_blank(),
        axis.line.x.top = element_blank(),
        axis.text.y = element_text(color = "white")) +
  # scale break is breaking coord_cartesian(lim)
  ↪ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
  scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↪ as.POSIXct(c("2021-05-01", "2023-11-01"))) +
  #coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle

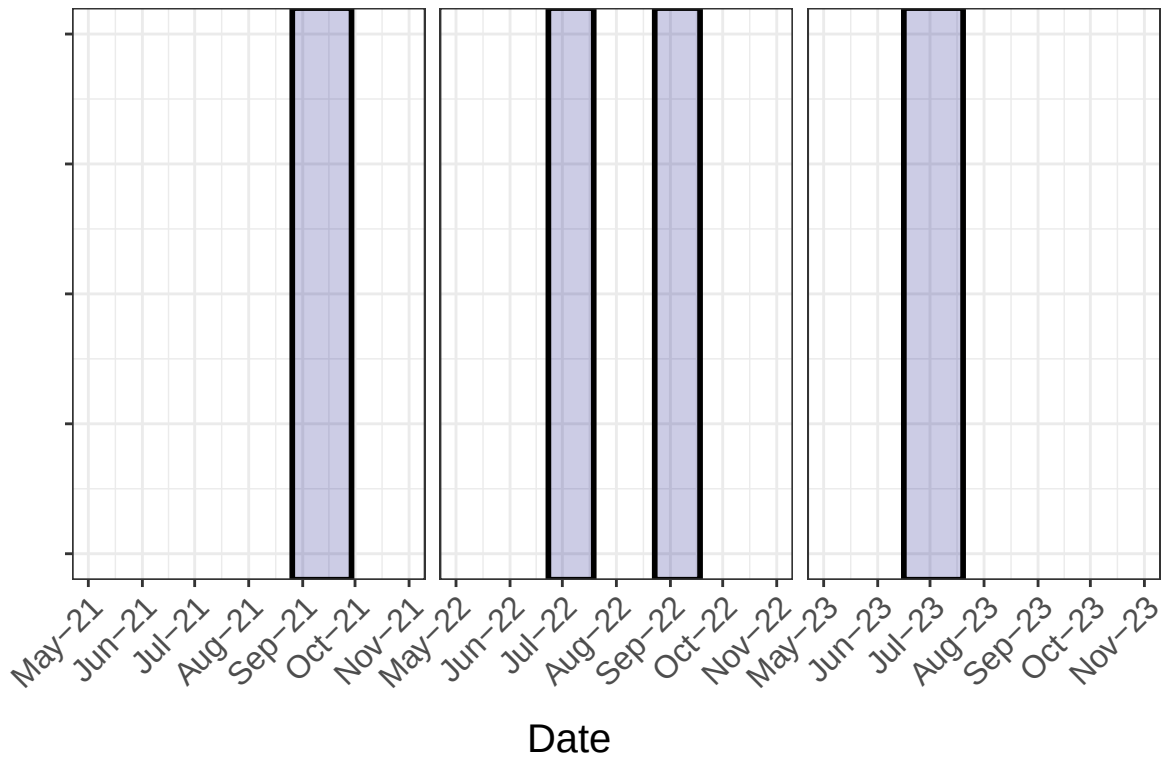
```

```

ylim(c(-0.5, 0.5)) +
# clip = "off" allows data points to be drawn in the margins
# expand = 0 doesn't add an expansion factor to the limits to keep the data off the
↪ axes
ggtitle("PPS Deployment Windows") +
xlab("Date") +
scale_x_break(as.POSIXct(c("2021-11-01", "2022-05-01"))) +
scale_x_break(as.POSIXct(c("2022-11-01", "2023-05-01")))

```

PPS Deployment Windows



```

ggsave(filename = here("Manuscript_Plots/DeploymentWindowsBoxes.png"),
        width = widthpx,
        height = heightpx,
        units = "px")

```

Stacked Years

```

# Graph 2: Stacked Years

plt1 <- ggplot() +
  SampHighlight2 + # in 2021
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 12),
        strip.background = element_rect(fill = "gray95"), # facet label background
        axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
        ↪ ggbreak

```

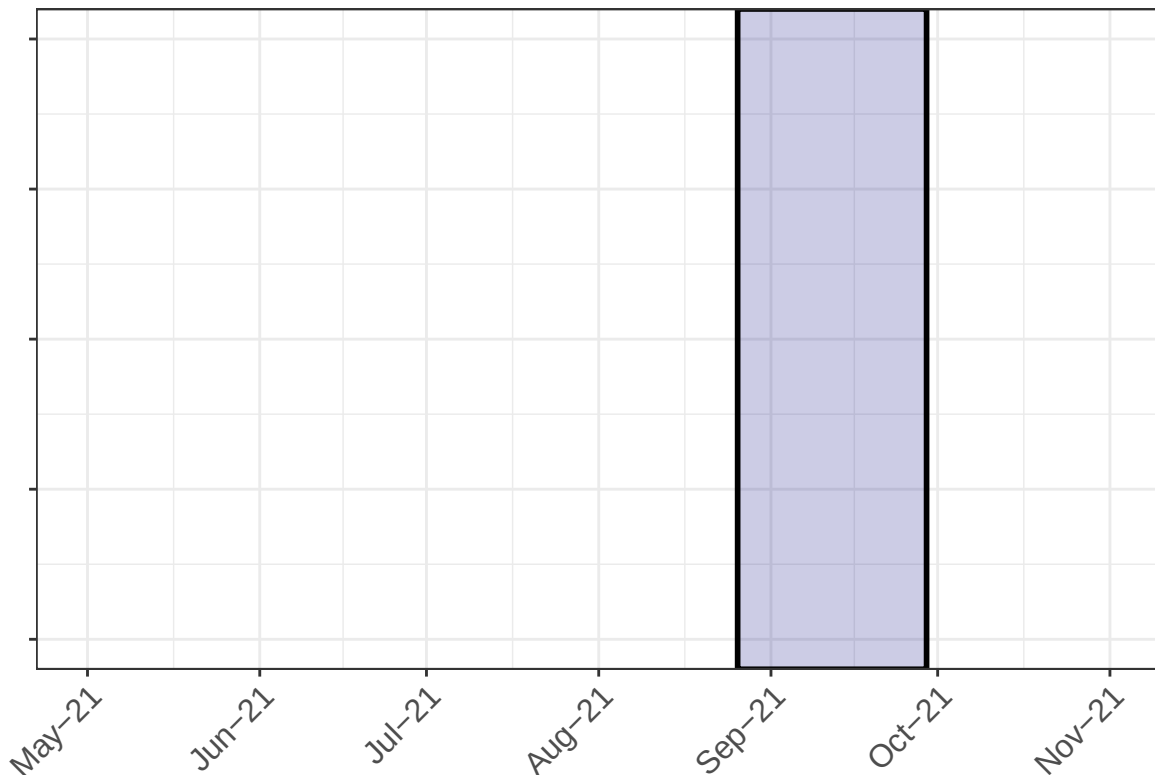
```

axis.ticks.x.top = element_blank(),
axis.line.x.top = element_blank(),
axis.text.y = element_text(color = "white")) +
# scale break is breaking coord_cartesian(lim)
→ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
→ as.POSIXct(c("2021-05-01", "2021-11-01"))) +
#coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
ylim(c(-0.5, 0.5)) +
# clip = "off" allows data points to be drawn in the margins
# expand = 0 doesn't add an expansion factor to the limits to keep the data off the
→ axes
ggtitle("PPS Deployment Windows")

```

plt1

PPS Deployment Windows



```

plt2 <- ggplot() +
  SampHighlight1 + # in 2022
  SampHighlight4 +
  theme_bw() +
  theme(text = element_text(size = 15),
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.text = element_text(size = 12),
    strip.background = element_rect(fill = "gray95"), # facet label background
    axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
    → ggbreak
    axis.ticks.x.top = element_blank(),

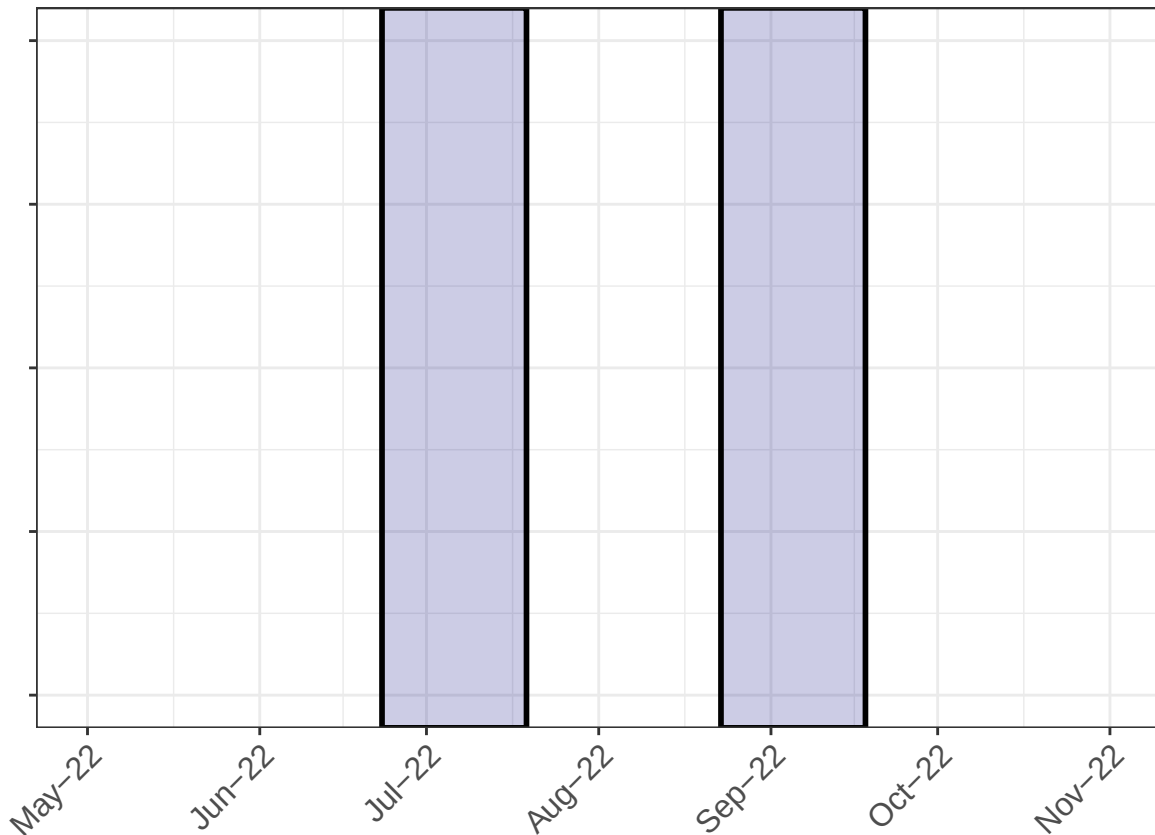
```

```

axis.line.x.top = element_blank(),
axis.text.y = element_text(color = "white")) +
# scale break is breaking coord_cartesian(lim)
↳ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
↳ as.POSIXct(c("2022-05-01", "2022-11-01"))) +
#coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
ylim(c(-0.5, 0.5))
# clip = "off" allows data points to be drawn in the margins
# expand = 0 doesn't add an expansion factor to the limits to keep the data off the
↳ axes

```

plt2



```

plt3 <- ggplot() +
  SampHighlight3 + # in 2023
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 12),
        strip.background = element_rect(fill = "gray95"), # facet label background
        axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
        ↳ ggbreak
        axis.ticks.x.top = element_blank(),
        axis.line.x.top = element_blank(),
        axis.text.y = element_text(color = "white")) +
# scale break is breaking coord_cartesian(lim)
↳ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before

```

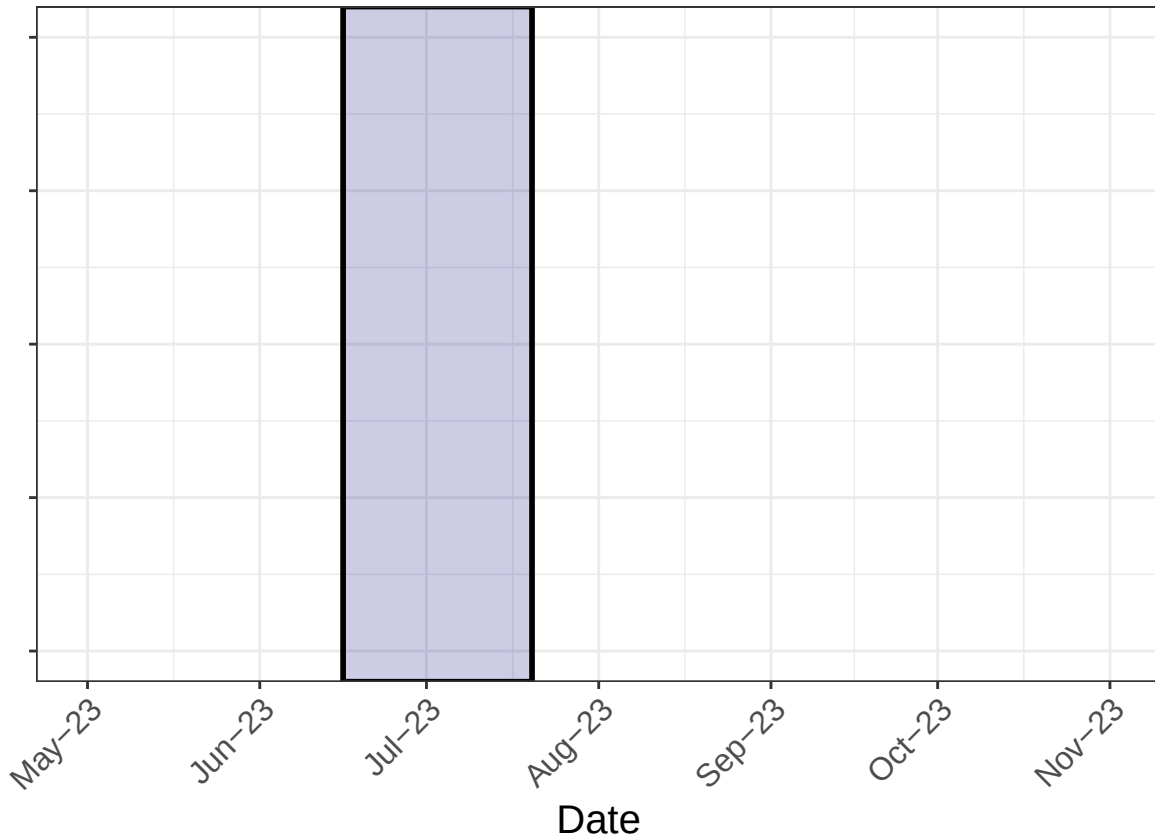


```

scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↳ as.POSIXct(c("2023-05-01", "2023-11-01"))) +
#coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
ylim(c(-0.5, 0.5)) +
# clip = "off" allows data points to be drawn in the margins
# expand = 0 doesn't add an expansion factor to the limits to keep the data off the
  ↳ axes
xlab("Date")

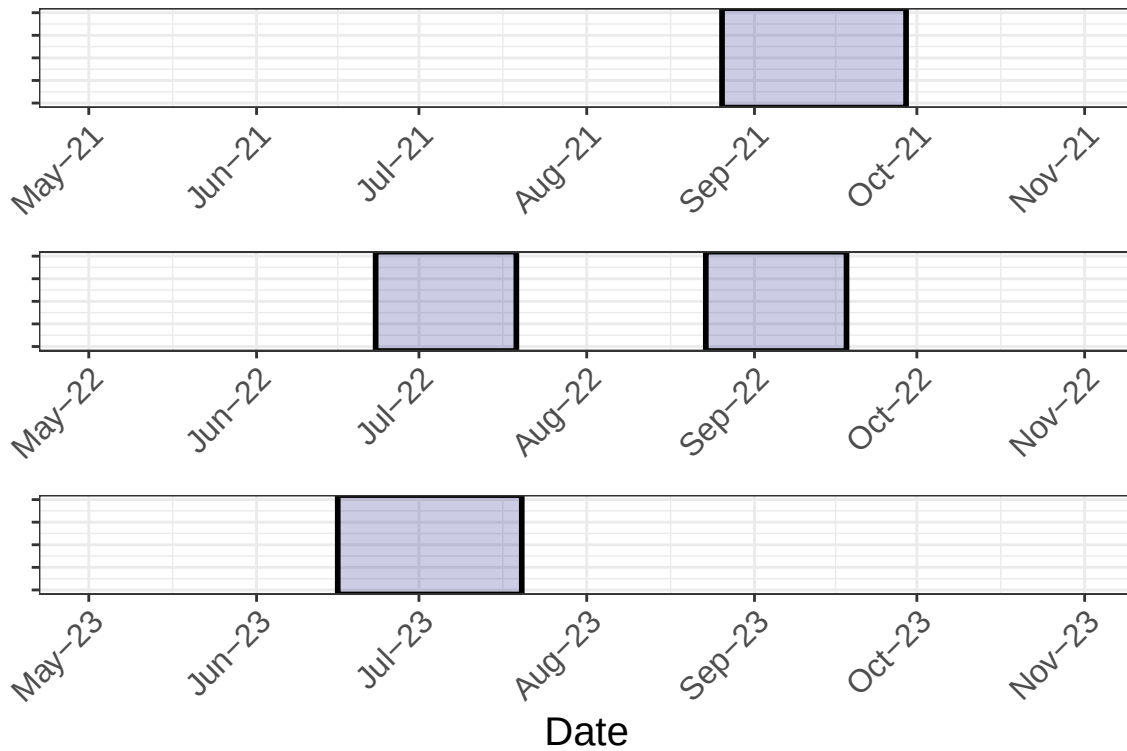
```

plt3



plt1/plt2/plt3

PPS Deployment Windows



```
ggsave(filename = here("Manuscript_Plots/DeploymentWindowsBoxesStacked.png"),
        width = widthpx_stacked,
        height = heightpx_stacked,
        units = "px")
```

Each Sample Marked

Graph 3: Each sample marked

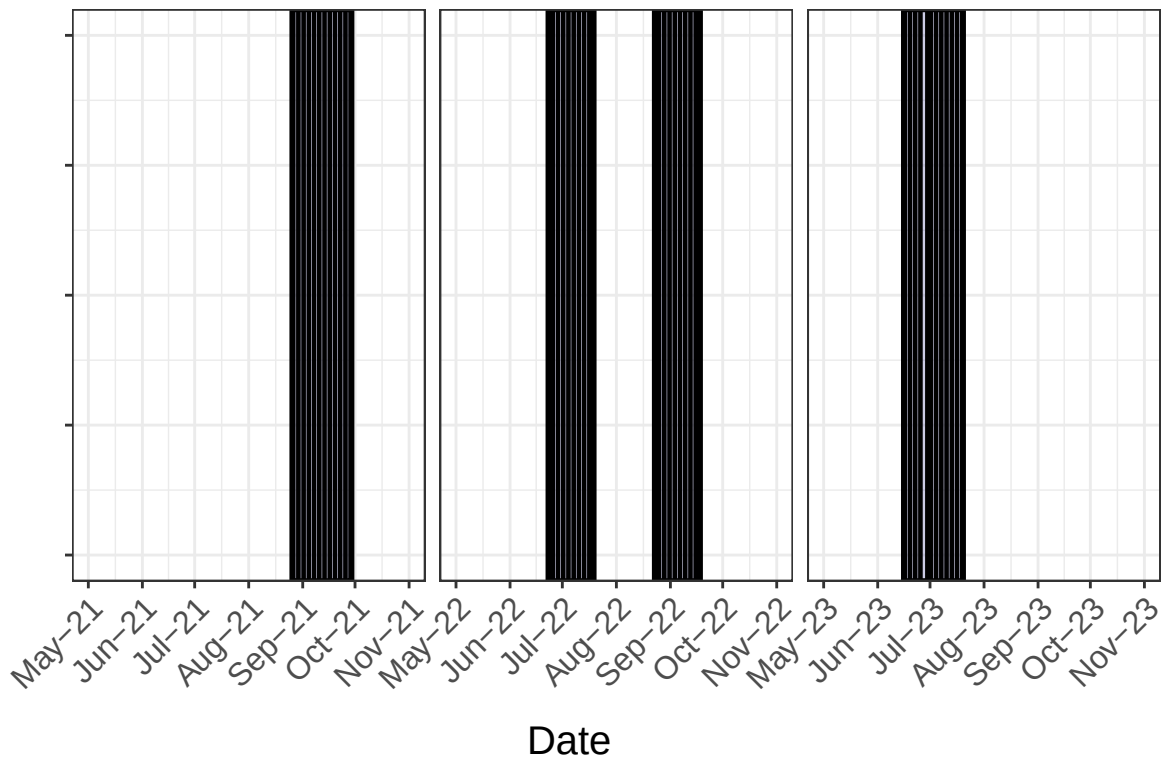
```
ggplot() +
  SampHighlight1 +
  SampHighlight2 +
  SampHighlight3 +
  SampHighlight4 +
  geom_vline(aes(xintercept = sampledatalist)) +
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 12),
        strip.background = element_rect(fill = "gray95"), # facet label background
        axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
        ↪ ggbreak
        axis.ticks.x.top = element_blank(),
        axis.line.x.top = element_blank(),
        axis.text.y = element_text(color = "white")) +
  # scale break is breaking coord_cartesian(lim)
  ↪ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
```

```

scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↳ as.POSIXct(c("2021-05-01", "2023-11-01"))) +
#coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
ylim(c(-0.5, 0.5)) +
# clip = "off" allows data points to be drawn in the margins
# expand = 0 doesn't add an expansion factor to the limits to keep the data off the
  ↳ axes
ggtitle("PPS Deployment Windows") +
xlab("Date") +
scale_x_break(as.POSIXct(c("2021-11-01", "2022-05-01"))) +
scale_x_break(as.POSIXct(c("2022-11-01", "2023-05-01")))

```

PPS Deployment Windows



```

ggsave(filename = here("Manuscript_Plots/DeploymentWindowsBoxesLines.png"),
  width = widthpx,
  height = heightpx,
  units = "px")

```

Stacked Years

```

# Graph 2: Stacked Years

plt1 <- ggplot() +
  SampHighlight2 + # in 2021
  geom_vline(aes(xintercept = sampledatalist)) +
  theme_bw() +
  theme(text = element_text(size = 15),

```

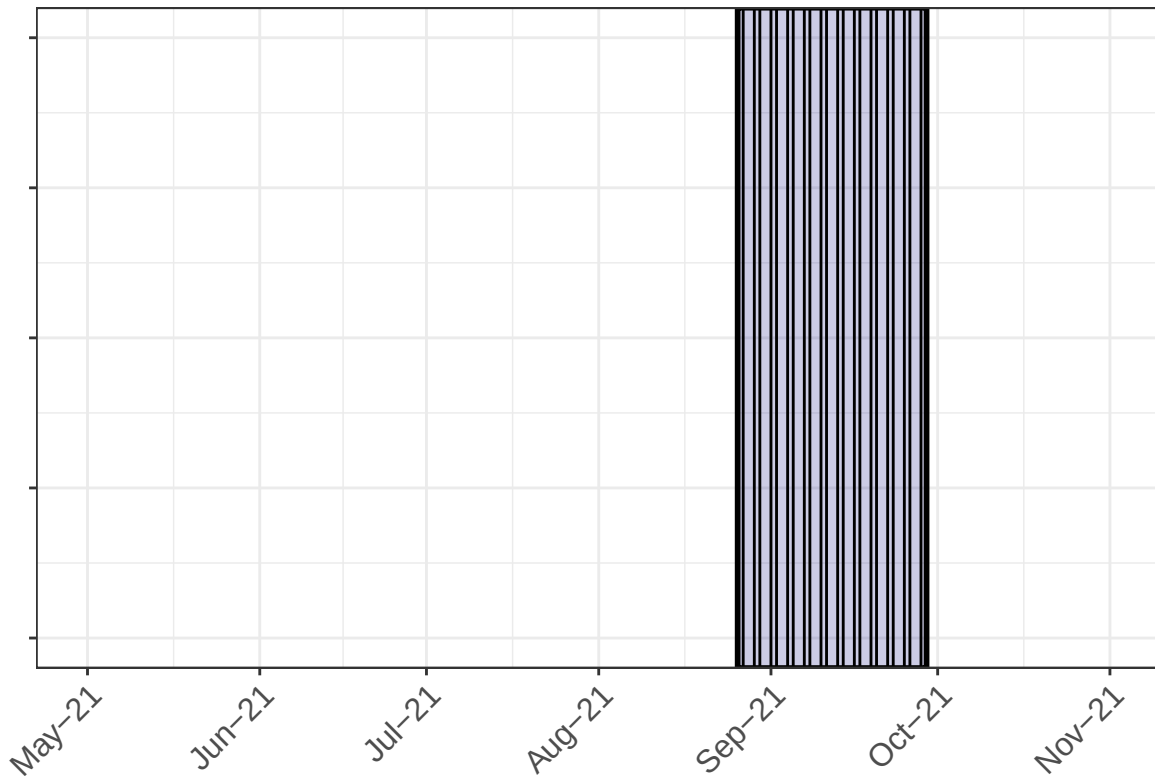
```

axis.text.x = element_text(angle = 45, hjust = 1),
strip.text = element_text(size = 12),
strip.background = element_rect(fill = "gray95"), # facet label background
axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
  ↳ ggbreak
axis.ticks.x.top = element_blank(),
axis.line.x.top = element_blank(),
axis.text.y = element_text(color = "white")) +
# scale break is breaking coord_cartesian(lim)
  ↳ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↳ as.POSIXct(c("2021-05-01", "2021-11-01"))) +
#coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
ylim(c(-0.5, 0.5)) +
# clip = "off" allows data points to be drawn in the margins
# expand = 0 doesn't add an expansion factor to the limits to keep the data off the
  ↳ axes
ggtitle("PPS Deployment Windows")

```

plt1

PPS Deployment Windows



```

plt2 <- ggplot() +
  SampHighlight1 + # in 2022
  SampHighlight4 +
  geom_vline(aes(xintercept = sampledatalist)) +
  theme_bw() +
  theme(text = element_text(size = 15),

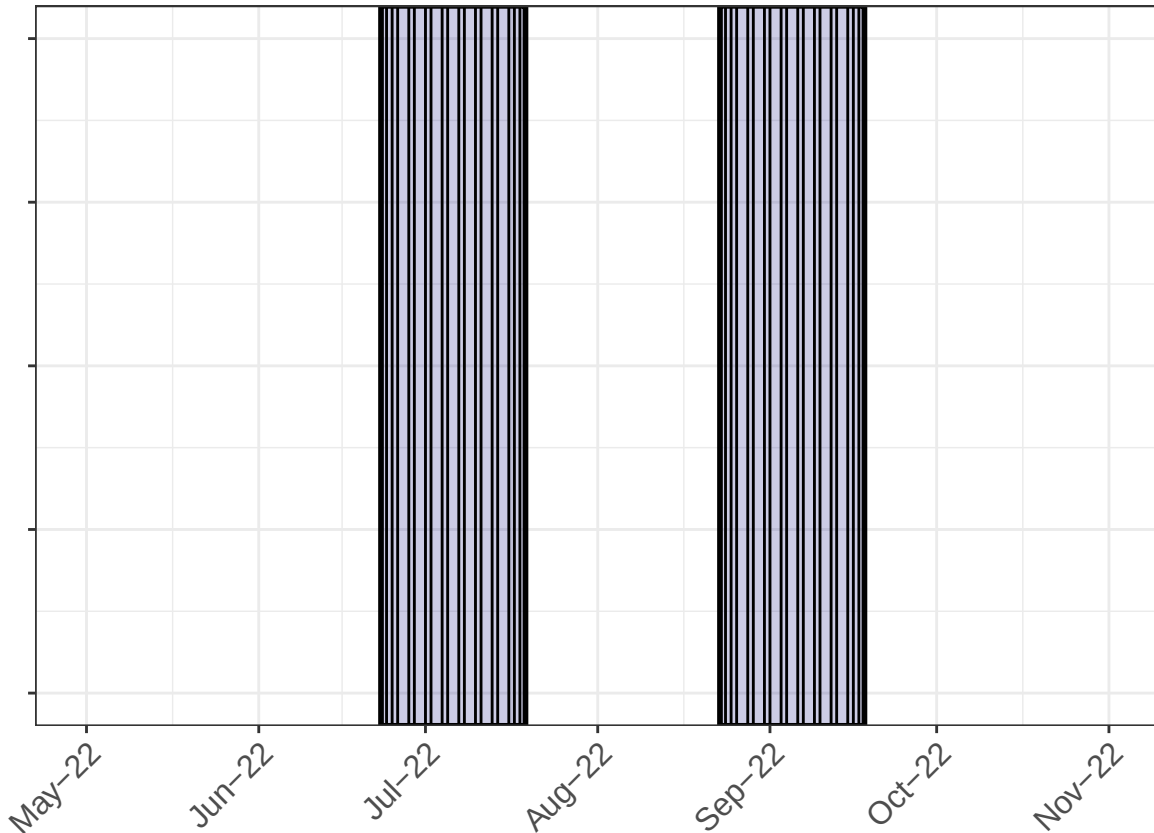
```

```

axis.text.x = element_text(angle = 45, hjust = 1),
strip.text = element_text(size = 12),
strip.background = element_rect(fill = "gray95"), # facet label background
axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
↳ ggbreak
axis.ticks.x.top = element_blank(),
axis.line.x.top = element_blank(),
axis.text.y = element_text(color = "white")) +
# scale break is breaking coord_cartesian(lim)
↳ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
↳ as.POSIXct(c("2022-05-01", "2022-11-01"))) +
#coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
ylim(c(-0.5, 0.5))
# clip = "off" allows data points to be drawn in the margins
# expand = 0 doesn't add an expansion factor to the limits to keep the data off the
↳ axes

```

plt2



```

plt3 <- ggplot() +
  SampHighlight3 + # in 2023
  geom_vline(aes(xintercept = sampledatelist)) +
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 12),

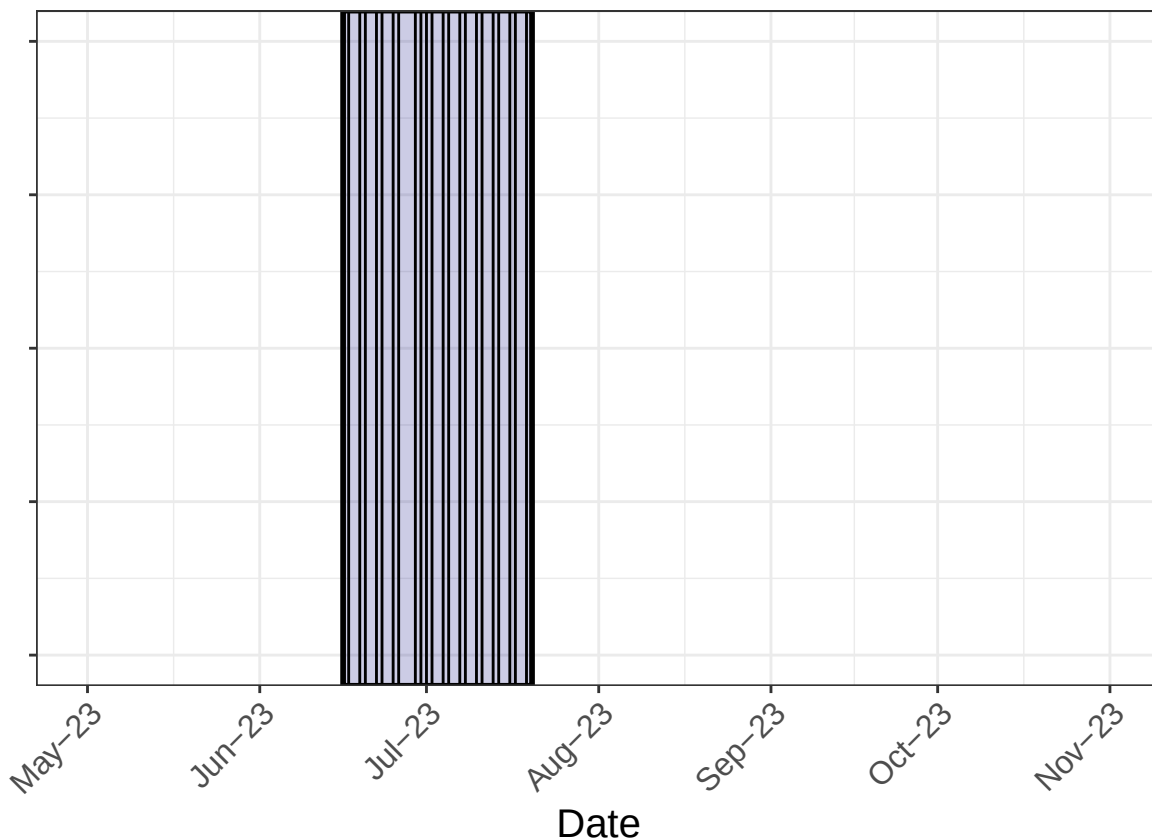
```

```

strip.background = element_rect(fill = "gray95"), # facet label background
axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
  ↳ ggbreak
axis.ticks.x.top = element_blank(),
axis.line.x.top = element_blank(),
axis.text.y = element_text(color = "white")) +
# scale break is breaking coord_cartesian(lim)
  ↳ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↳ as.POSIXct(c("2023-05-01", "2023-11-01"))) +
#coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
ylim(c(-0.5, 0.5)) +
# clip = "off" allows data points to be drawn in the margins
# expand = 0 doesn't add an expansion factor to the limits to keep the data off the
  ↳ axes
xlab("Date")

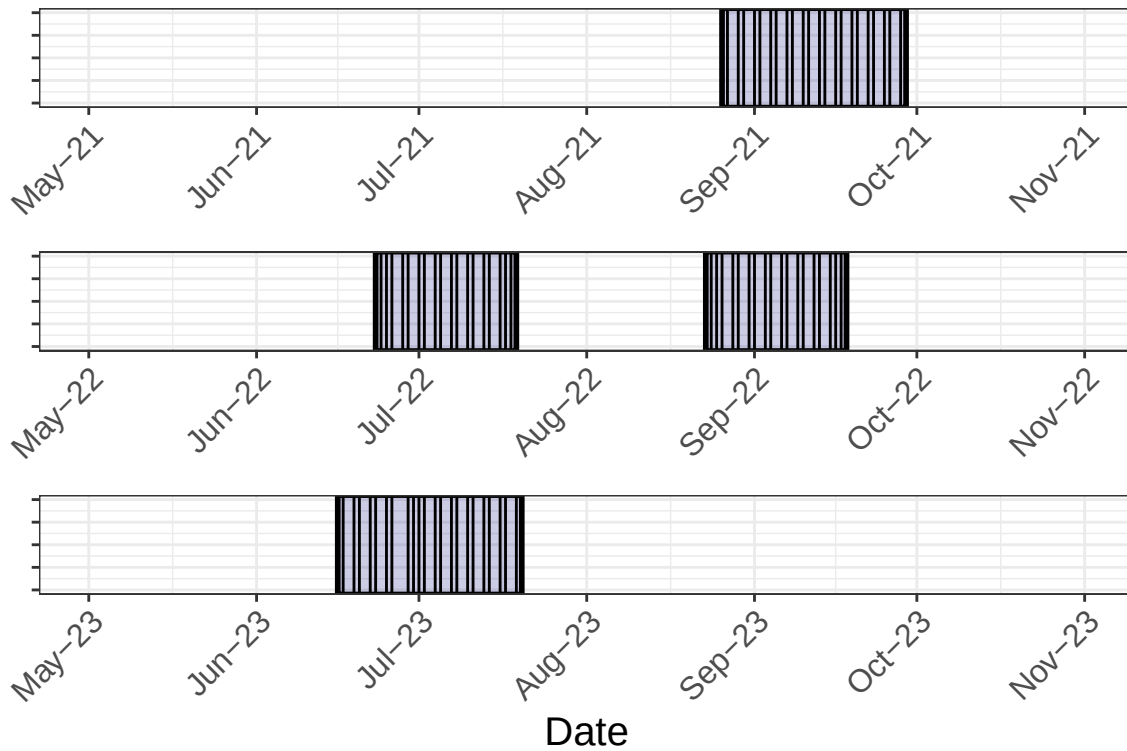
```

plt3



plt1/plt2/plt3

PPS Deployment Windows



```
ggsave(filename = here("Manuscript_Plots/DeploymentWindowsBoxesLinesStacked.png"),
        width = widthpx_stacked,
        height = heightpx_stacked,
        units = "px")
```

CTD + geom_rect

Graph 1: Shaded gray sampling windows

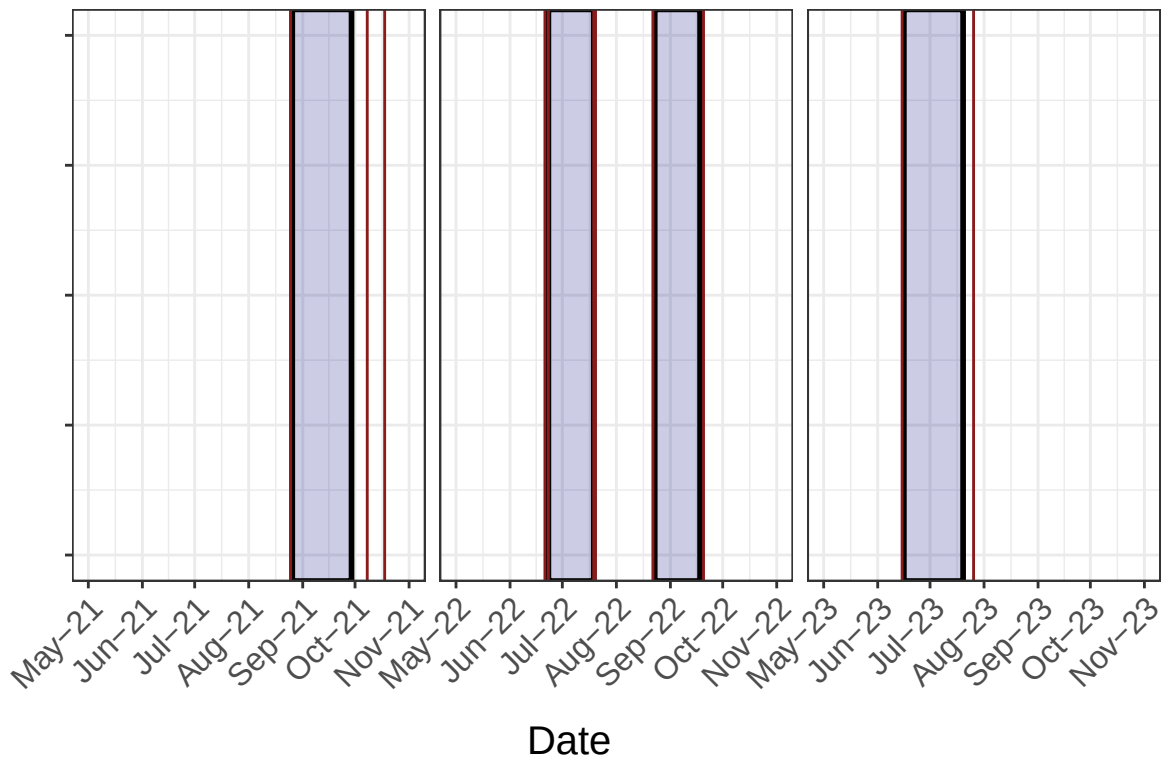
```
ggplot() +
  SampHighlight1 +
  SampHighlight2 +
  SampHighlight3 +
  SampHighlight4 +
  geom_vline(aes(xintercept = ctddatelist), color = ctdcolor) + # add CTD
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 12),
        strip.background = element_rect(fill = "gray95"), # facet label background
        axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
        ↪ ggbreak
        axis.ticks.x.top = element_blank(),
        axis.line.x.top = element_blank(),
        axis.text.y = element_text(color = "white")) +
  # scale break is breaking coord_cartesian(lim)
  ↪ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
```

```

scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↳ as.POSIXct(c("2021-05-01", "2023-11-01"))) +
#coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
ylim(c(-0.5, 0.5)) +
# clip = "off" allows data points to be drawn in the margins
# expand = 0 doesn't add an expansion factor to the limits to keep the data off the
  ↳ axes
ggtitle("PPS Deployment Windows") +
xlab("Date") +
scale_x_break(as.POSIXct(c("2021-11-01", "2022-05-01"))) +
scale_x_break(as.POSIXct(c("2022-11-01", "2023-05-01")))

```

PPS Deployment Windows



```

ggsave(filename = here("Manuscript_Plots/DeploymentWindowsBoxesCTD.png"),
  width = widthpx,
  height = heightpx,
  units = "px")

```

CTD as dots

```

# Graph 1: Shaded gray sampling windows

CTD_dots <- geom_point(aes(x = ctddatelist, y = 0), color = ctdcolor) # make CTDs dots to
  ↳ distinguish, since they're usually on the borders of deployment windows

ggplot() +
  SampHighlight1 +

```

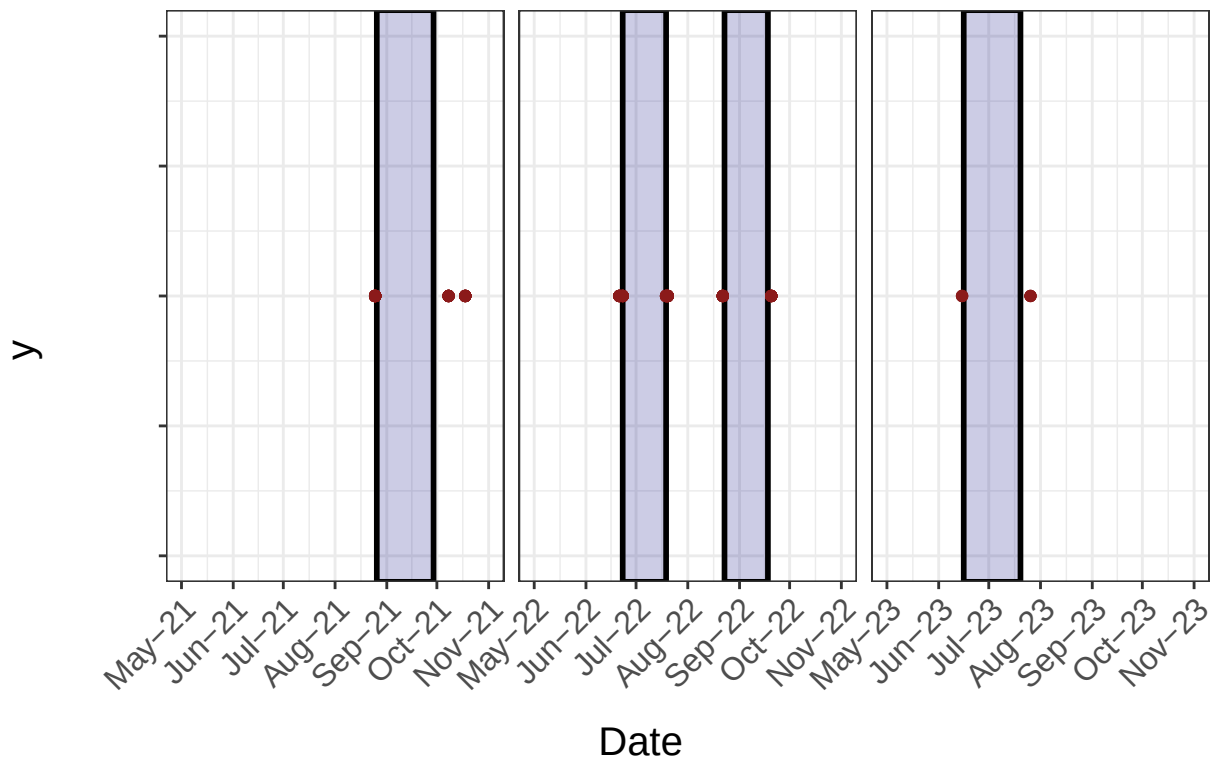


```

SampHighlight2 +
SampHighlight3 +
SampHighlight4 +
CTD_dots +
theme_bw() +
theme(text = element_text(size = 15),
      axis.text.x = element_text(angle = 45, hjust = 1),
      strip.text = element_text(size = 12),
      strip.background = element_rect(fill = "gray95"), # facet label background
      axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
      ↪ ggbreak
      axis.ticks.x.top = element_blank(),
      axis.line.x.top = element_blank(),
      axis.text.y = element_text(color = "white")) +
# scale break is breaking coord_cartesian(lim)
↪ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
↪ as.POSIXct(c("2021-05-01", "2023-11-01"))) +
#coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
ylim(c(-0.5, 0.5)) +
# clip = "off" allows data points to be drawn in the margins
# expand = 0 doesn't add an expansion factor to the limits to keep the data off the
↪ axes
ggtitle("PPS Deployment Windows") +
xlab("Date") +
scale_x_break(as.POSIXct(c("2021-11-01", "2022-05-01"))) +
scale_x_break(as.POSIXct(c("2022-11-01", "2023-05-01")))

```

PPS Deployment Windows



```
ggsave(filename = here("Manuscript_Plots/DeploymentWindowsBoxesCTDDots.png"),
        width = widthpx,
        height = heightpx,
        units = "px")
```

CTD + Stacked Years

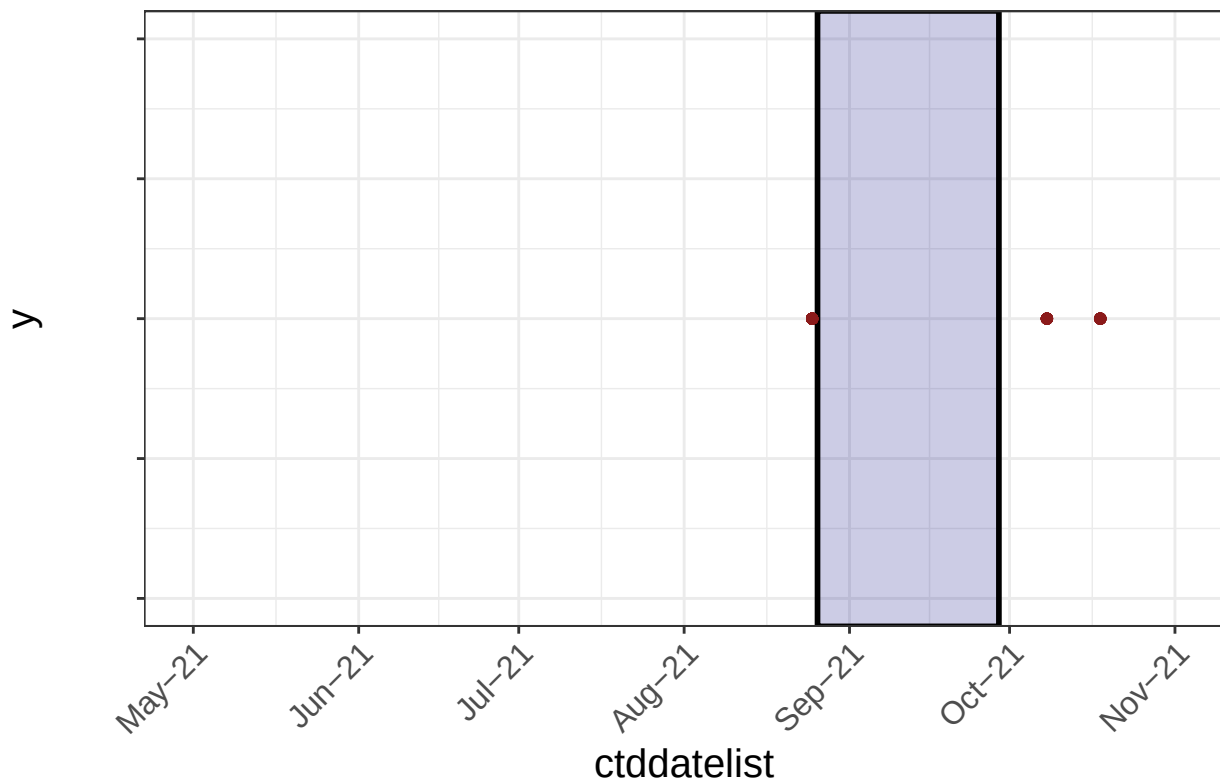
Graph 2: Stacked Years

```
plt1 <- ggplot() +
  SampHighlight2 + # in 2021
  CTD_add +
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 12),
        strip.background = element_rect(fill = "gray95"), # facet label background
        axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
        ↪ ggbreak
        axis.ticks.x.top = element_blank(),
        axis.line.x.top = element_blank(),
        axis.text.y = element_text(color = "white")) +
  # scale break is breaking coord_cartesian(lim)
  ↪ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
  scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↪ as.POSIXct(c("2021-05-01", "2021-11-01"))) +
  #coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
  ylim(c(-0.5, 0.5)) +
  # clip = "off" allows data points to be drawn in the margins
  # expand = 0 doesn't add an expansion factor to the limits to keep the data off the
  ↪ axes
  ggtitle("PPS Deployment Windows")

plt1
```

```
## Warning: Removed 99 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

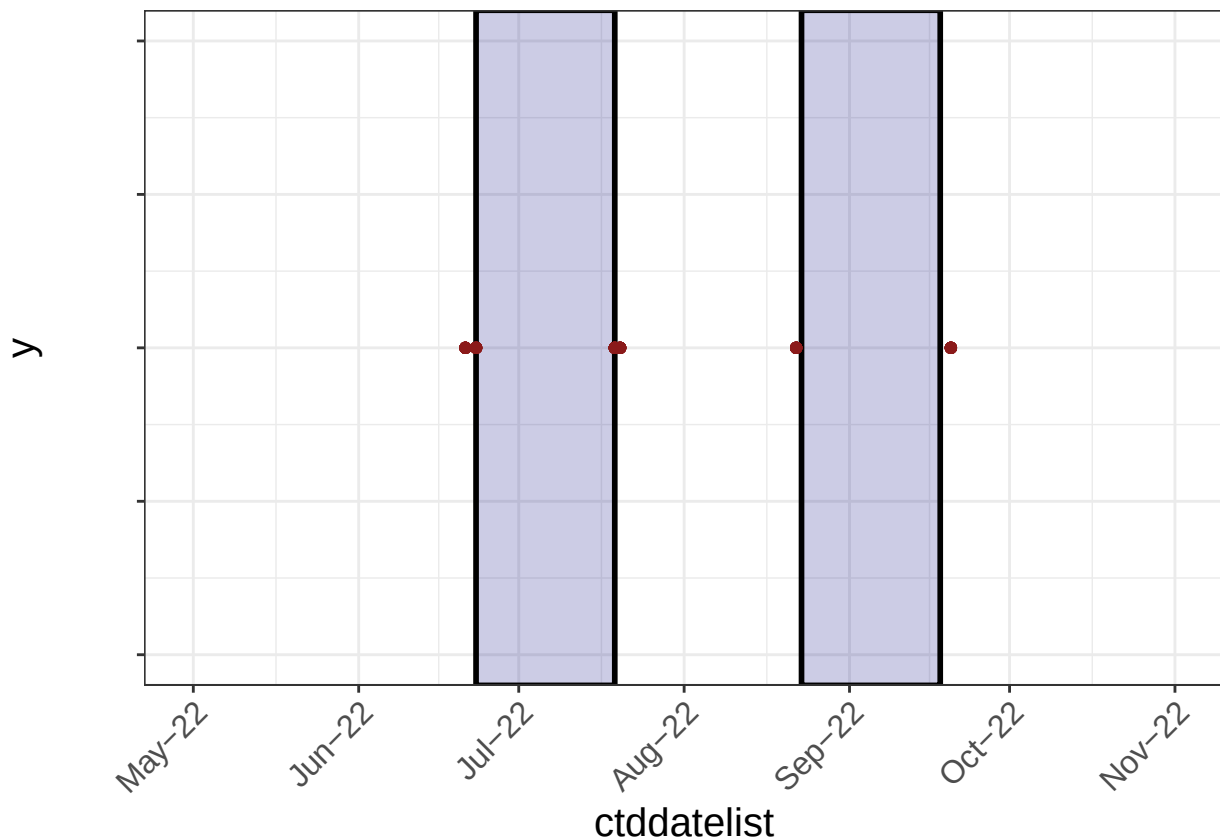
PPS Deployment Windows



```
plt2 <- ggplot() +
  SampHighlight1 + # in 2022
  SampHighlight4 +
  CTD_add +
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 12),
        strip.background = element_rect(fill = "gray95"), # facet label background
        axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
        ↪ ggbreak
        axis.ticks.x.top = element_blank(),
        axis.line.x.top = element_blank(),
        axis.text.y = element_text(color = "white")) +
  # scale break is breaking coord_cartesian(lim)
  ↪ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
  scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↪ as.POSIXct(c("2022-05-01", "2022-11-01"))) +
  #coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
  ylim(c(-0.5, 0.5))
  # clip = "off" allows data points to be drawn in the margins
  # expand = 0 doesn't add an expansion factor to the limits to keep the data off the
  ↪ axes

plt2
```

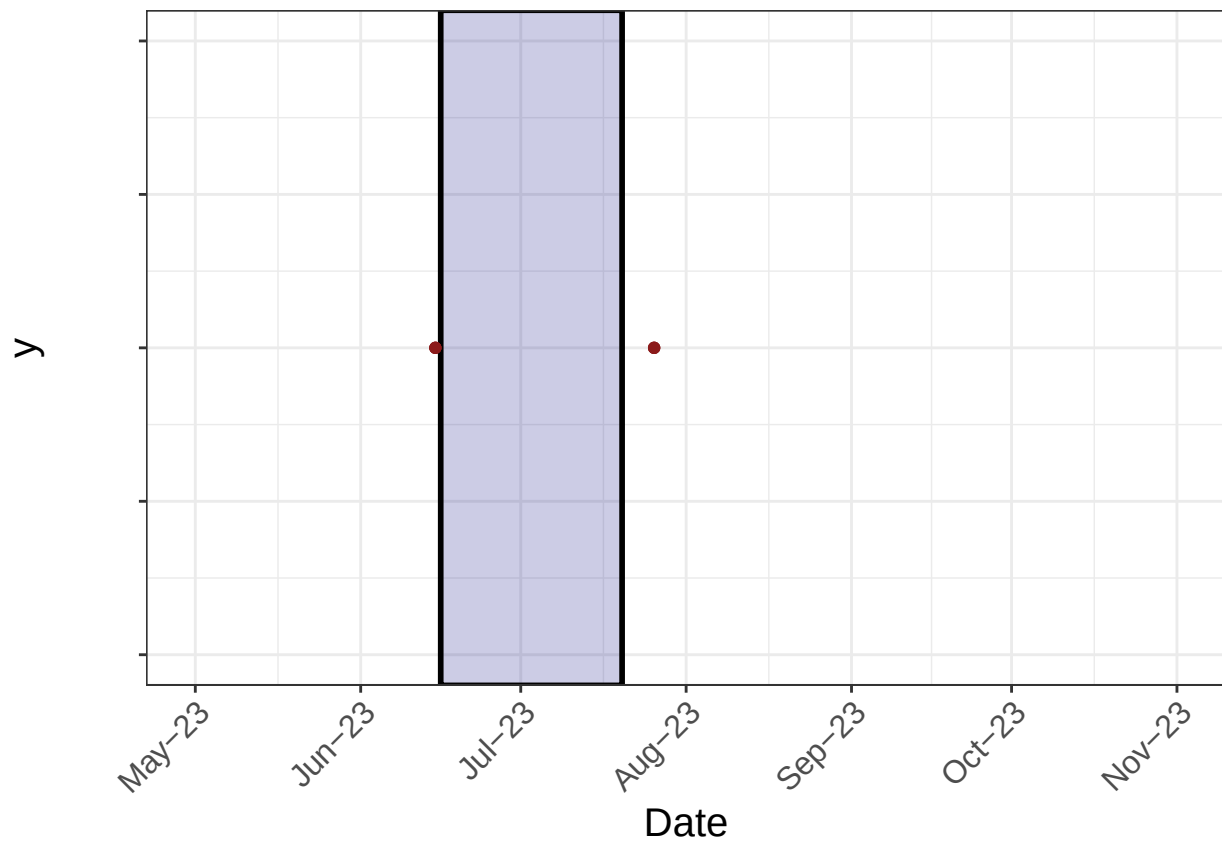
```
## Warning: Removed 77 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



```
plt3 <- ggplot() +
  SampHighlight3 + # in 2023
  CTD_add +
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 12),
        strip.background = element_rect(fill = "gray95"), # facet label background
        axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
        ↪ ggbreak
        axis.ticks.x.top = element_blank(),
        axis.line.x.top = element_blank(),
        axis.text.y = element_text(color = "white")) +
  # scale break is breaking coord_cartesian(lim)
  ↪ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
  scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↪ as.POSIXct(c("2023-05-01", "2023-11-01"))) +
  #coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
  ylim(c(-0.5, 0.5)) +
  # clip = "off" allows data points to be drawn in the margins
  # expand = 0 doesn't add an expansion factor to the limits to keep the data off the
  ↪ axes
  xlab("Date")
```

```
plt3
```

```
## Warning: Removed 137 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



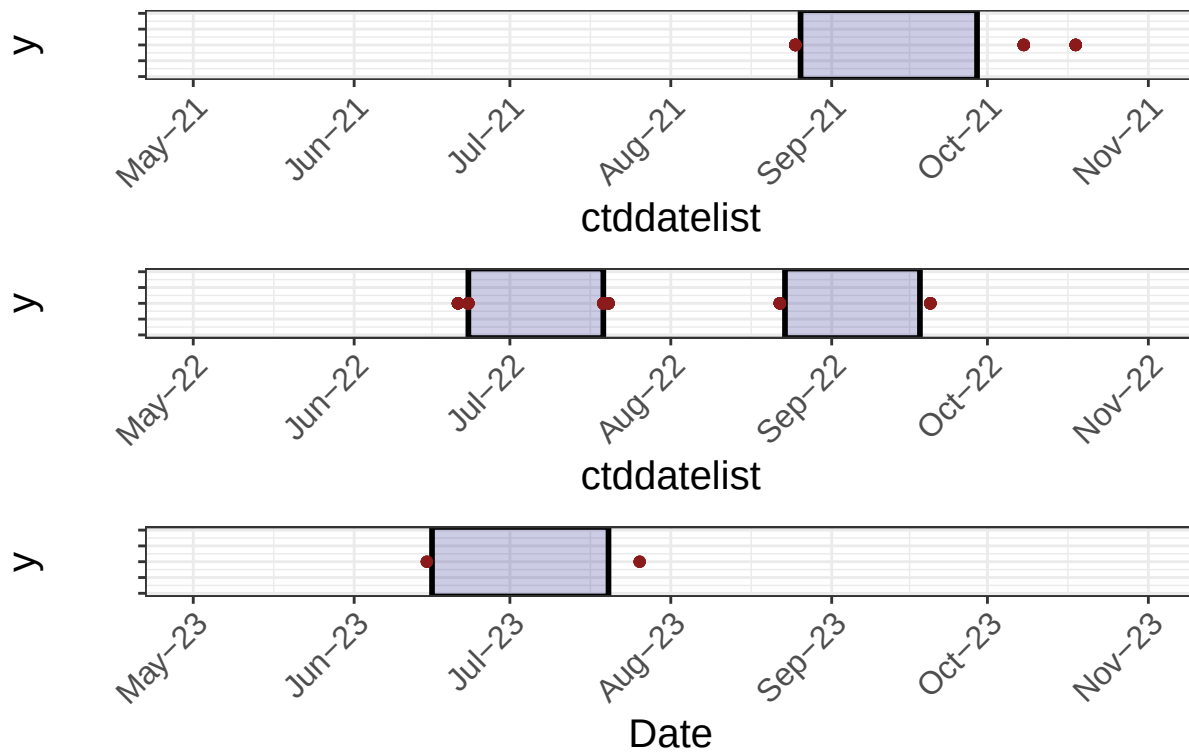
```
plt1/plt2/plt3
```

```
## Warning: Removed 99 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 77 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 137 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

PPS Deployment Windows



```
ggsave(filename = here("Manuscript_Plots/DeploymentWindowsBoxesStackedCTD.png"),
        width = widthpx_stacked,
        height = heightpx_stacked,
        units = "px")
```

```
## Warning: Removed 99 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 77 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 137 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

CTD + Each Sample Marked

```
# Graph 3: Each sample marked
```

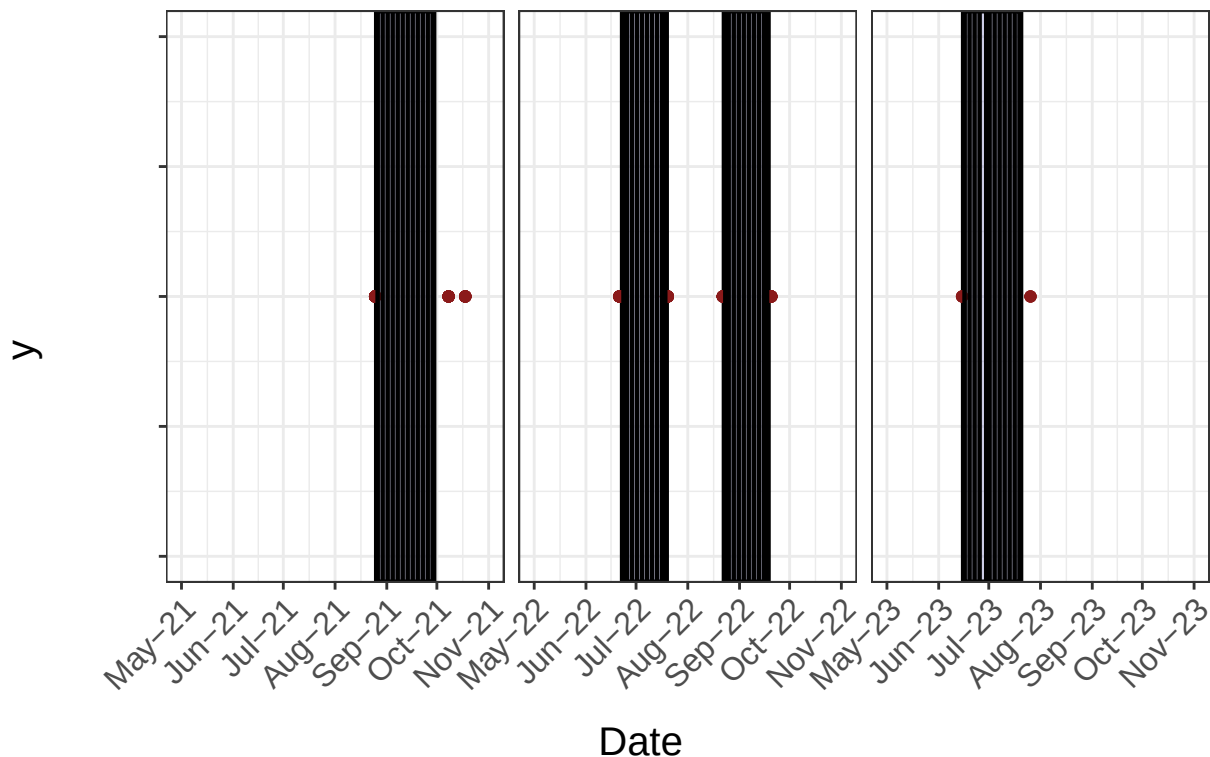
```
ggplot() +
  SampHighlight1 +
  SampHighlight2 +
  SampHighlight3 +
  SampHighlight4 +
  CTD_add +
  geom_vline(aes(xintercept = sampleditelist)) +
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
```

```

strip.text = element_text(size = 12),
strip.background = element_rect(fill = "gray95"), # facet label background
axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
  ↳ ggbreak
axis.ticks.x.top = element_blank(),
axis.line.x.top = element_blank(),
axis.text.y = element_text(color = "white")) +
# scale break is breaking coord_cartesian(lim)
  ↳ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↳ as.POSIXct(c("2021-05-01", "2023-11-01"))) +
#coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
ylim(c(-0.5, 0.5)) +
# clip = "off" allows data points to be drawn in the margins
# expand = 0 doesn't add an expansion factor to the limits to keep the data off the
  ↳ axes
ggtitle("PPS Deployment Windows") +
xlab("Date") +
scale_x_break(as.POSIXct(c("2021-11-01", "2022-05-01"))) +
scale_x_break(as.POSIXct(c("2022-11-01", "2023-05-01")))

```

PPS Deployment Windows



```

ggsave(filename = here("Manuscript_Plots/DeploymentWindowsBoxesLinesCTD.png"),
        width = widthpx,
        height = heightpx,
        units = "px")

```

CTD + Stacked Years

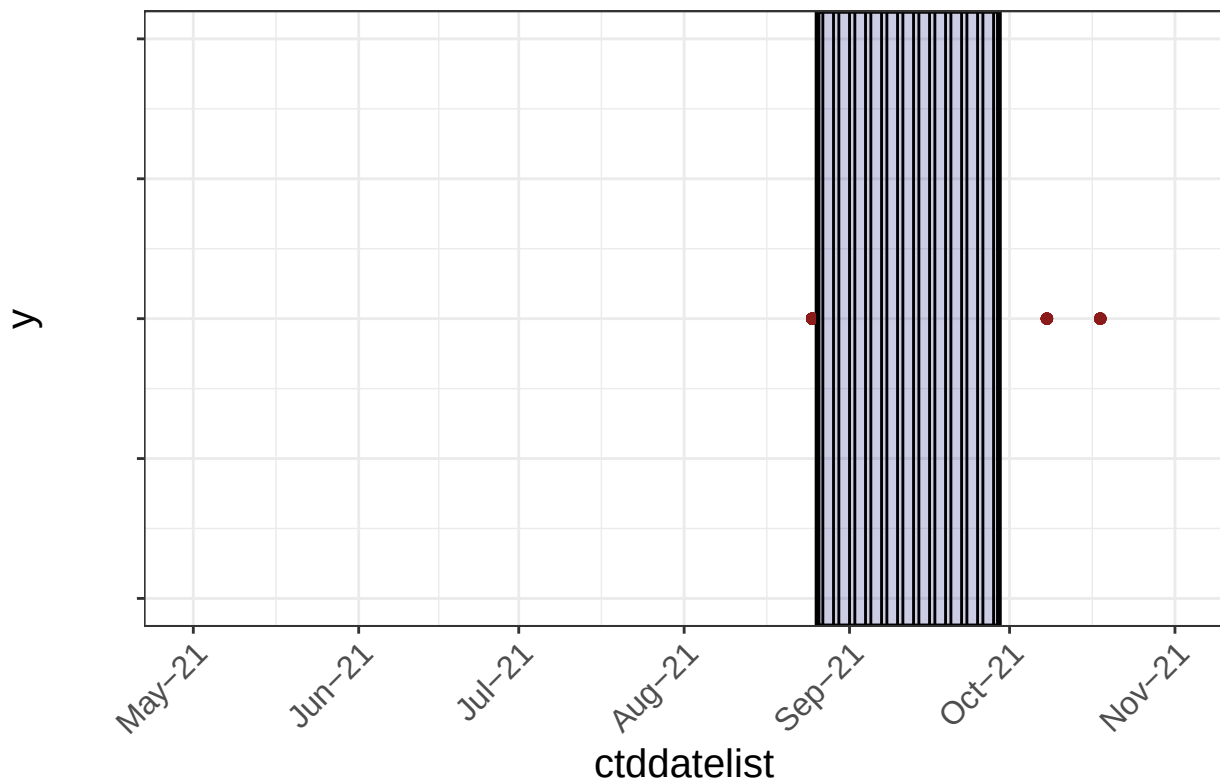
```
# Graph 2: Stacked Years

plt1 <- ggplot() +
  SampHighlight2 + # in 2021
  CTD_add +
  geom_vline(aes(xintercept = sampledatelist)) +
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 12),
        strip.background = element_rect(fill = "gray95"), # facet label background
        axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
        ↪ ggbreak
        axis.ticks.x.top = element_blank(),
        axis.line.x.top = element_blank(),
        axis.text.y = element_text(color = "white")) +
  # scale break is breaking coord_cartesian(lim)
  ↪ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
  scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↪ as.POSIXct(c("2021-05-01", "2021-11-01"))) +
  #coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
  ylim(c(-0.5, 0.5)) +
  # clip = "off" allows data points to be drawn in the margins
  # expand = 0 doesn't add an expansion factor to the limits to keep the data off the
  ↪ axes
  ggtitle("PPS Deployment Windows")

plt1
```

```
## Warning: Removed 99 rows containing missing values or values outside the scale range
## (`geom_point()`).
```


PPS Deployment Windows

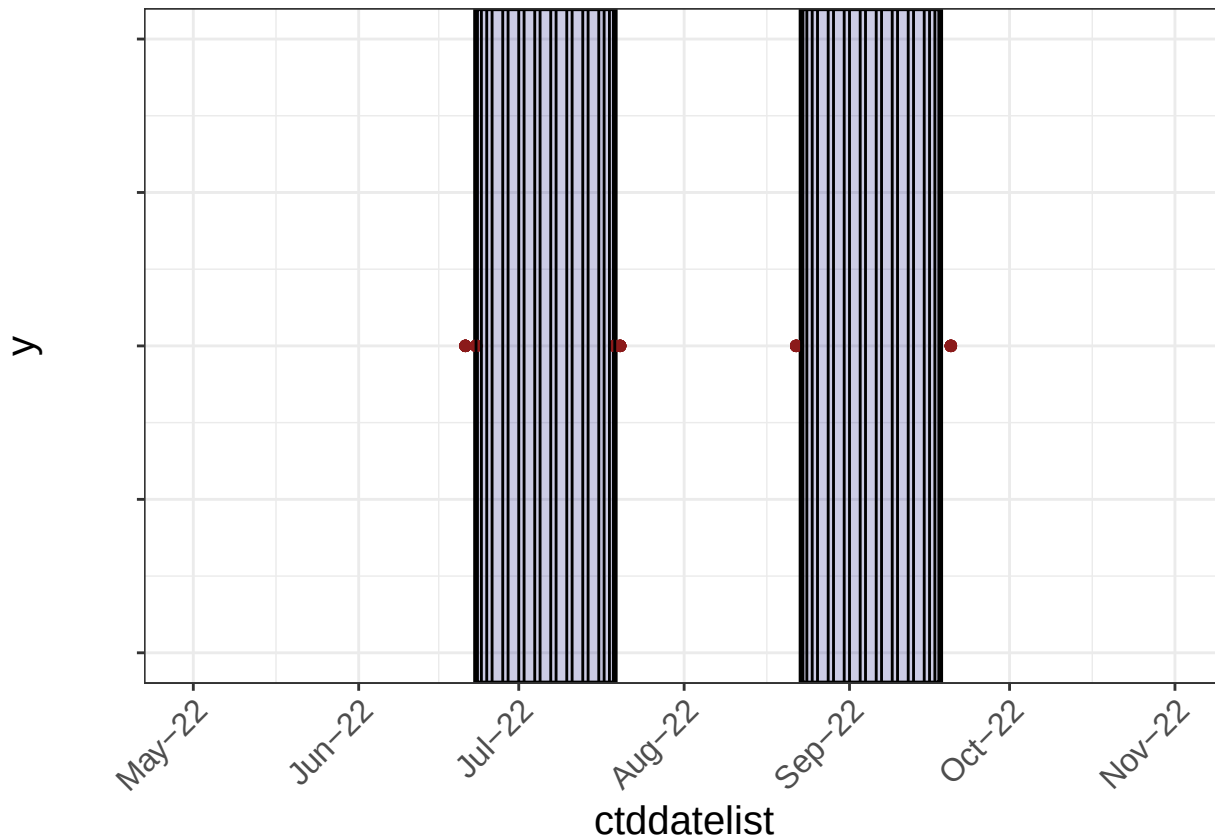


```
plt2 <- ggplot() +
  SampHighlight1 + # in 2022
  SampHighlight4 +
  CTD_add +
  geom_vline(aes(xintercept = sampledatelist)) +
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 12),
        strip.background = element_rect(fill = "gray95"), # facet label background
        axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
        ↪ ggbreak
        axis.ticks.x.top = element_blank(),
        axis.line.x.top = element_blank(),
        axis.text.y = element_text(color = "white")) +
  # scale break is breaking coord_cartesian(lim)
  ↪ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
  scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↪ as.POSIXct(c("2022-05-01", "2022-11-01"))) +
  #coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
  ylim(c(-0.5, 0.5))
  # clip = "off" allows data points to be drawn in the margins
  # expand = 0 doesn't add an expansion factor to the limits to keep the data off the
  ↪ axes

plt2
```

Warning: Removed 77 rows containing missing values or values outside the scale range

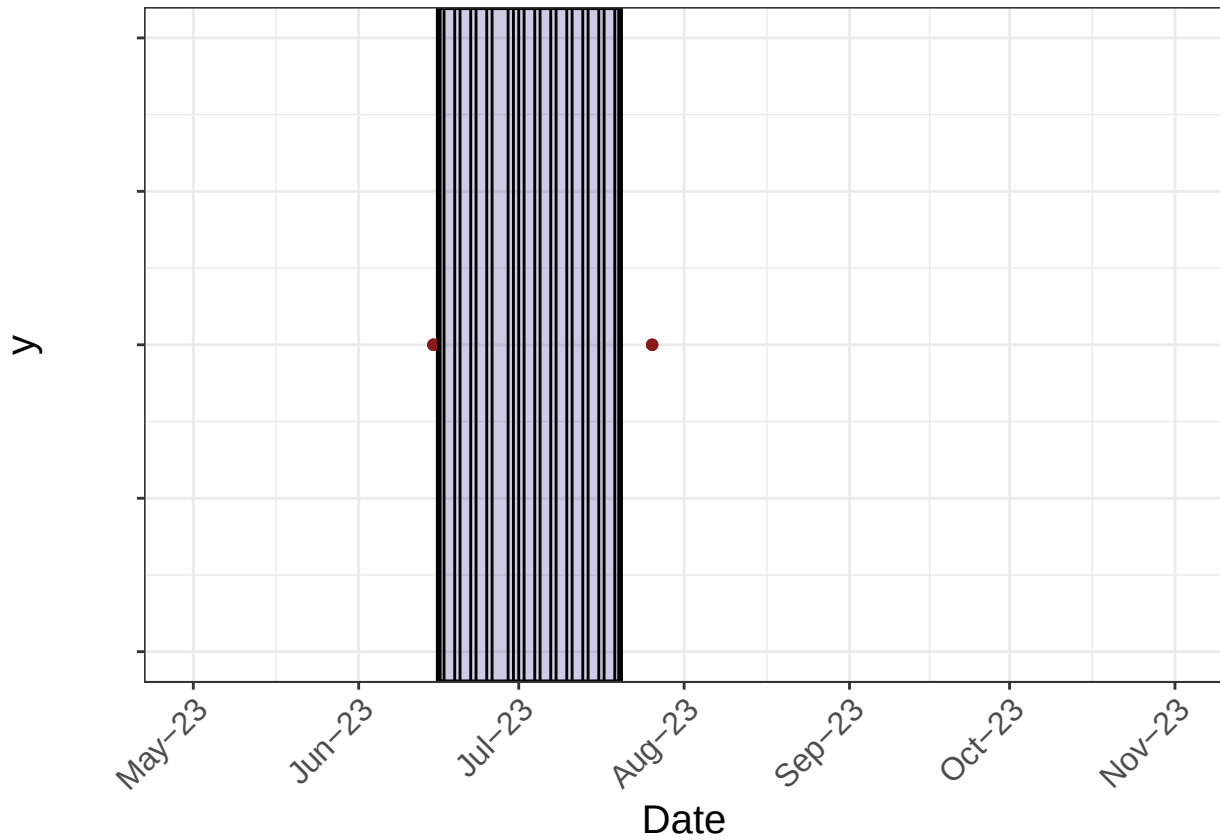
```
## (`geom_point()`).
```



```
plt3 <- ggplot() +
  SampHighlight3 + # in 2023
  CTD_add +
  geom_vline(aes(xintercept = sampledatelist)) +
  theme_bw() +
  theme(text = element_text(size = 15),
        axis.text.x = element_text(angle = 45, hjust = 1),
        strip.text = element_text(size = 12),
        strip.background = element_rect(fill = "gray95"), # facet label background
        axis.text.x.top = element_blank(), # Needed to delete the extra axis created by
        ↪ ggbreak
        axis.ticks.x.top = element_blank(),
        axis.line.x.top = element_blank(),
        axis.text.y = element_text(color = "white")) +
  # scale break is breaking coord_cartesian(lim)
  ↪ https://github.com/YuLab-SMU/ggbreak/issues/59 put cartesian before
  scale_x_datetime(breaks = "month", date_labels = "%b-%y", limits =
  ↪ as.POSIXct(c("2023-05-01", "2023-11-01"))) +
  #coord_cartesian(ylim = c(-0.5, 0.5)) + # Zooms in the plot without deleting rectangle
  ylim(c(-0.5, 0.5)) +
  # clip = "off" allows data points to be drawn in the margins
  # expand = 0 doesn't add an expansion factor to the limits to keep the data off the
  ↪ axes
  xlab("Date")
```

```
plt3
```

```
## Warning: Removed 137 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



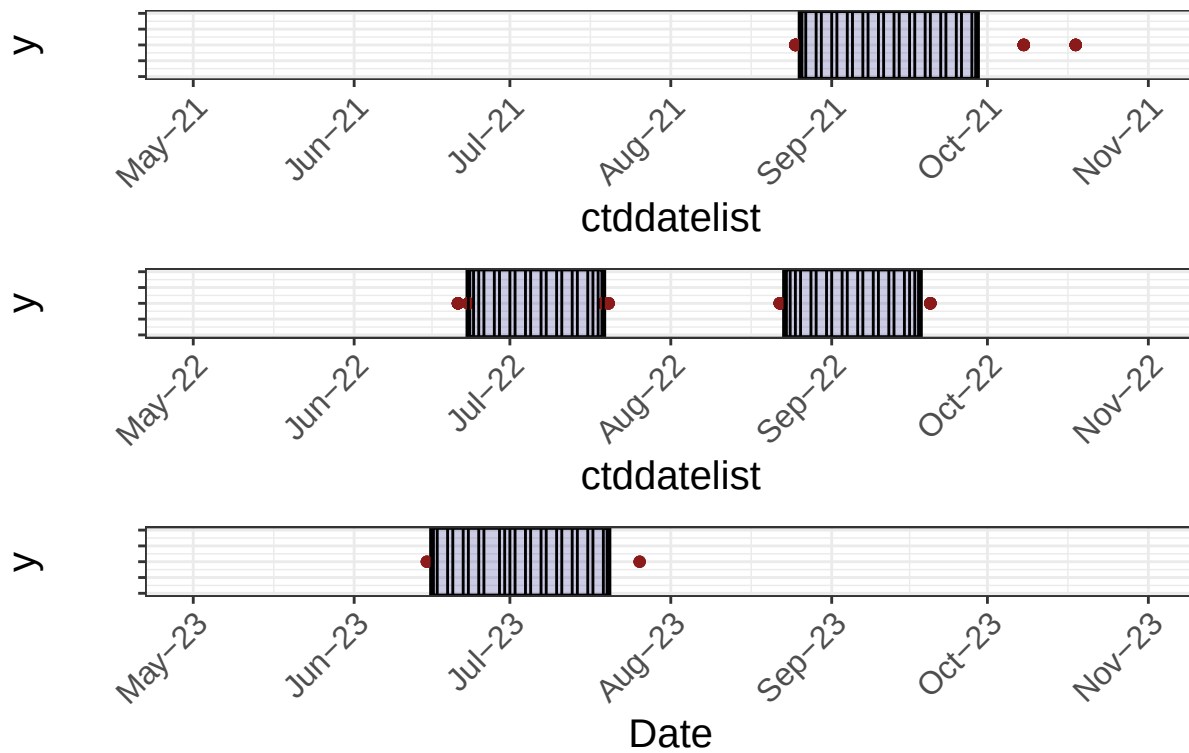
```
plt1/plt2/plt3
```

```
## Warning: Removed 99 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 77 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 137 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

PPS Deployment Windows



```
ggsave(filename = here("Manuscript_Plots/DeploymentWindowsBoxesLinesStackedCTD.png"),
        width = widthpx_stacked,
        height = heightpx_stacked,
        units = "px")
```

```
## Warning: Removed 99 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 77 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 137 rows containing missing values or values outside the scale range
## (`geom_point()`).
```